

Assignment: File Handling in C Programming

1. Create playlist.txt and Write Top 3 Favorite Songs

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    FILE *fp;
```

```
    fp = fopen("playlist.txt", "w");
```

```
    fprintf(fp, "Believer\n");
```

```
    fprintf(fp, "Tum Hi Ho\n");
```

```
    fprintf(fp, "Blinding Lights\n");
```

```
    fclose(fp);
```

```
    printf("Songs written successfully.\n");
```

```
    return 0;
```

```
}
```

Contents of playlist.txt

Believer

Tum Hi Ho

Blinding Lights

2. Read playlist.txt and Display Each Song

```
#include <stdio.h>

int main()
{
    FILE *fp;
    char song[100];

    fp = fopen("playlist.txt", "r");

    printf("Songs in Playlist:\n");

    while(fgets(song, sizeof(song), fp) != NULL)
    {
        printf("%s", song);
    }

    fclose(fp);

    return 0;
}
```

Sample Output

Songs in Playlist:

Believer

Tum Hi Ho

Blinding Lights

3. Append Two More Songs to playlist.txt

```
#include <stdio.h>

int main()
{
    FILE *fp;

    fp = fopen("playlist.txt", "a");

    fprintf(fp, "Love Story\n");
    fprintf(fp, "Crazy in Love\n");

    fclose(fp);

    printf("Songs appended successfully.\n");

    return 0;
}
```

Updated playlist.txt

Believer
Tum Hi Ho
Blinding Lights
Love Story
Crazy in Love

4. Read and Print Only Songs Containing "love" (Case-Insensitive)

```
#include <stdio.h>

#include <string.h>

#include <ctype.h>


int main()
{
    FILE *fp;

    char song[100];
    char temp[100];


    fp = fopen("playlist.txt", "r");


    printf("Songs containing 'love':\n");


    while(fgets(song, sizeof(song), fp) != NULL)
    {
        strcpy(temp, song);


        for(int i = 0; temp[i]; i++)
        {
            temp[i] = tolower(temp[i]);
        }


        if(strstr(temp, "love") != NULL)
        {
            printf("%s", song);
```

```
        }  
    }  
  
    fclose(fp);  
  
    return 0;  
}
```

Sample Output

Songs containing 'love':

Love Story

Crazy in Love

Conclusion

This assignment demonstrates:

- Creating files using Write Mode (w)
- Reading files using Read Mode (r)
- Adding data using Append Mode (a)
- Processing file contents using loops
- Searching text using string functions
- Performing case-insensitive filtering in C

These file handling techniques are essential for storing and retrieving data in real-world applications.